Howard Sun | Software Engineer

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SKILLS

TypeScript, JavaScript (ES6), React, Redux, Node, Express, PostgreSQL, MongoDB, HTML5, CSS (Sass), Tailwind, Material UI, Git, Docker, Kubernetes, GitHub Actions, WebSocket, AWS (EC2, S3, Aurora, Elastic Beanstalk, VPC), webpack, Vite, Solid.js, Jest, Playwright, Mocha, Chai, Electron.js, Python

EXPERIENCE

Swell · Software Engineer — April 2023 – Present

- Integrated Content Security Policy (CSP) into Swell for testing backend protocols using nonce and SHA-256 hashing, enhancing security against cross-site scripting (XSS) and data injection through defense-in-depth, protecting user data, and ensuring stable performance
- Managed tech debt by refactoring legacy Redux: Implemented type-safe hooks and Redux Toolkit, expanded TypeScript coverage, eliminated outdated functions, ensured consistent state management, and replaced ad-hoc type definitions, improving maintainability and code quality
- Developed a React-based binary file upload feature following Test-Driven Development (TDD) principles with Jest, enabling users to test backend server routes for diverse file types and sizes, expanding API testing capabilities when developing backend
- Boosted overall test coverage by 15% by repairing existing tests and writing new unit and integration tests using Jest, Mocha, Chai, and Playwright; enhanced test robustness with Jest's mocking functionality and Chai's semantic assertions
- Strengthened the reliability of Swell by identifying and resolving security vulnerabilities in the API Simulation Tool through root cause analysis and Node.js debugging, preventing application crashes and ensuring stability
- Standardized CSS and Material UI themes across the application, unifying CSS classes with Material UI theming and implementing dark mode, ensuring a consistent user experience

Mission Coders, LLC · CTO, Co-Founder — May 2018 – December 2022

- Transitioned from an on-premise monolithic server to a scalable AWS infrastructure with Elastic Load Balancing (ELB-ALB) and Elastic Compute Cloud (EC2), reducing resource overhead by 40%, enabling auto-scaling based on traffic patterns, and improving reliability and cost-efficiency across multiple regions
- Designed and implemented a scalable database solution using Amazon Aurora RDS to manage student and course registration; utilized Aurora's auto-scaling capabilities and read replicas to accommodate fluctuating traffic and high read workloads, ensuring a seamless user experience and 99% availability
- Led a six-member team to develop a children's programming learning platform using JavaScript, Node.js, Express, TypeScript, and Redux; integrated Scratch to deliver an engaging, interactive educational tool, completing the project 5 weeks ahead of the deadline

Quark Virtual Services · Technical Manager, Co-Founder — April 2013 – October 2022

- Streamlined deployment for managed and unmanaged Virtual Private Servers (VPS) using KVM and OpenVZ; applied infrastructure-as-code principles and configuration management (Ansible, Puppet) to create reusable, version-controlled templates, reducing manual effort and overall deployment time by 15%
- Executed SEO strategy by conducting keyword research, optimized on-page elements (title tags, meta descriptions, header tags, content), and establish off-page backlink campaign driving 35% organic traffic growth
- Increased conversion rate by 5% through SEO efforts by monitoring performance via Google Analytics, contributing to revenue growth and business objectives

Hewlett-Packard Enterprise · Technology Consultant II — January 2010 – November 2012

- Streamlined Federal Desktop Core Configuration (FDCC) rollouts for 500+ users by automating deployment using Bash and Python scripts; minimizing errors and ensured consistent configurations across all systems, saving time and resources while improving the overall efficiency and reliability of the deployment process
- Authored detailed electronic guides for hundreds of remote users to set up and troubleshoot workstation issues independently, reducing the infrastructure team's support burden by 800 hours
- Architected virtual environment testing processes prior to production server rollout, reducing software conflict-related downtime by 90%

PROJECTS

NestEgg – Personal Budgeting – Engineering Project

- Constructed a secure client-side routing system using React Router and PrivateRoute components, improving SPA navigation and safeguarding private budget entry pages through authentication checks

Home Automation – Engineering Project

- Engineered a home automation system using Raspberry Pi and Arduino, programmed in C++ for low-level control and performance. Integrated subsystems like automatic blind control with precise timing and PWM. Optimized memory and code execution for smooth operation, increasing home energy efficiency by 10%

EDUCATION

Central Connecticut State University – B.S. Electronic Technology

Kyoto University – B.S. Electrical and Electronics Engineering

INTERESTS

Diving into Korean and Russian languages; geeking out over all things Sci-Fi, especially Dune or Star Trek; exploring new cities and embracing nature while camping